

# **NE585**

## **NUCLEAR FUEL CYCLES**

### **Nuclear engineering basics**

# **1**

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# Learning objectives

Demonstrating fundamental concepts in nuclear physics

Analyzing decay data to identify radionuclides

Summarizing different decay mechanisms

Book – Chapter 2

# Learning nodes

**Relativity**

**Atomic structure**

**Binding energy**

**Q-value**

**Radioactivity**

**Chart of the nuclides**

**Decay chains**

**Half life**

# Relativity

# We apply energy from nuclear reactions and radiation interactions with matter

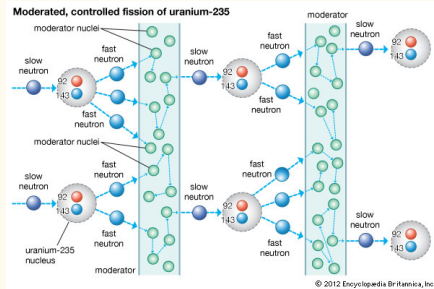
$$E_{REST} = m_0 c^2 \quad (1)$$

$$E = mc^2 \quad (2)$$

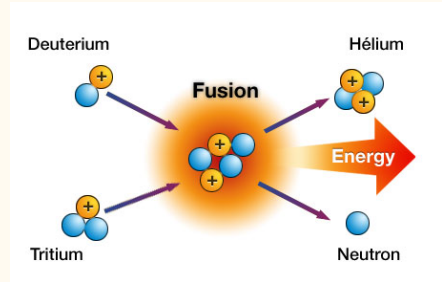
Rest mass energy for an electron – 0.511 *MeV*

Save that for later

# Energy released or energy injected



**Figure 1. Fission**



**Figure 2. Fusion**

# Determine the energy of a fusion neutron that is not at rest

$$m = \frac{m_0}{\sqrt{1 - (\frac{v^2}{c^2})}} \rightarrow E = mc^2 \quad (3)$$

With  $v = 5.2 \times 10^7 \text{ m/s}$ , that is about  $0.17c$ ; not negligible

So we have to compute the not rest mass

$$m_0(n) = 1.674927211 \times 10^{-27} \text{ kg}$$

Fusion neutron at that speed has  $E = 14.5 \text{ MeV}$

For relativistic calculations use MeV

# Atomic structure



# The shell model describes how subatomic particles are arranged

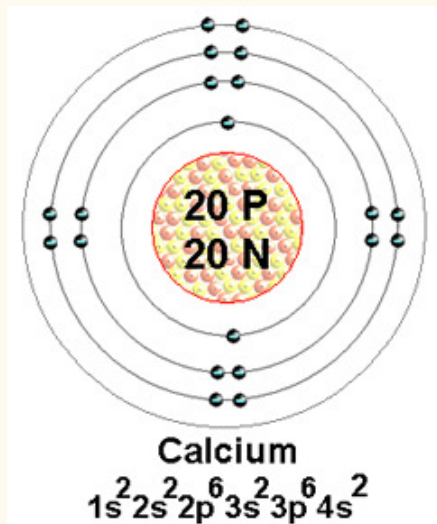


Figure 3. Calcium shell model

15: Phosphorus

2,8,5

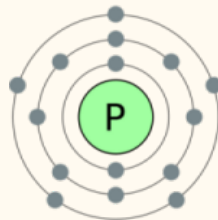


Figure 4. Phosphorus shell model

# Notation

$${}^A_Z Q_{A-Z} = {}^{235}_{92} U_{143} \quad (4)$$

# Periodic table of the elements

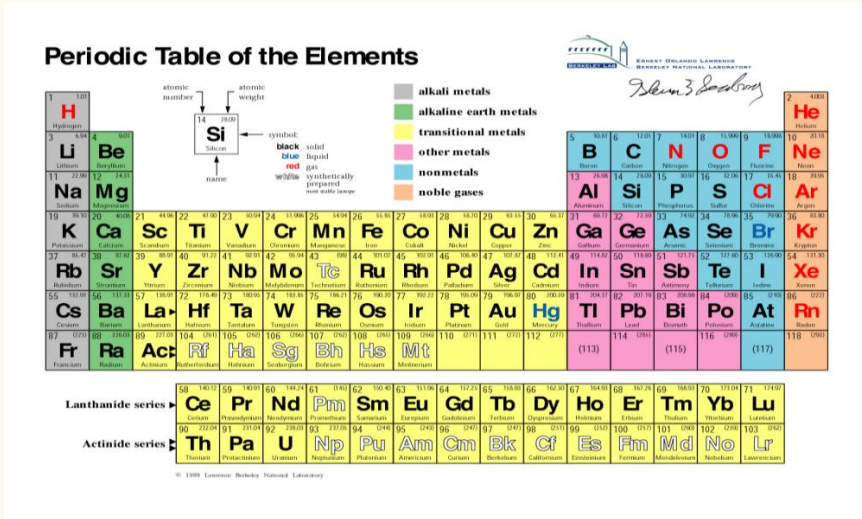


Figure 5. Periodic table of the elements

# Binding energy

# There are four fundamental forces of interaction

Gravity – Obvious

*Strong nuclear – Holds the nucleus together*

Electromagnetic – Basically electrons & magnetism

Weak nuclear – Changes the flavor of quarks

# To blow out a nucleus, a lot of energy is needed

Would it be constant or variable per nucleus?

Why would you want to blow out the nucleus?

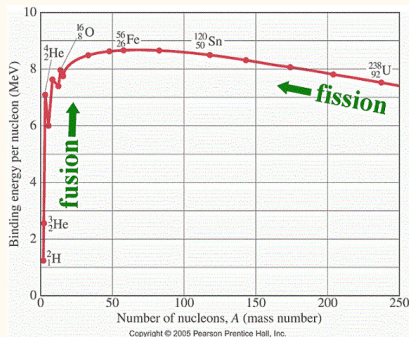
$$\Delta m = (A - Z)m_n - Zm_p - M \quad (5)$$

$$\Delta m({}^{239}_{94}\text{Pu}) = 1759 \text{ MeV} = 7.40 \frac{\text{MeV}}{\text{nucleon}} \quad (6)$$

This energy corresponds to a velocity of 0.88c

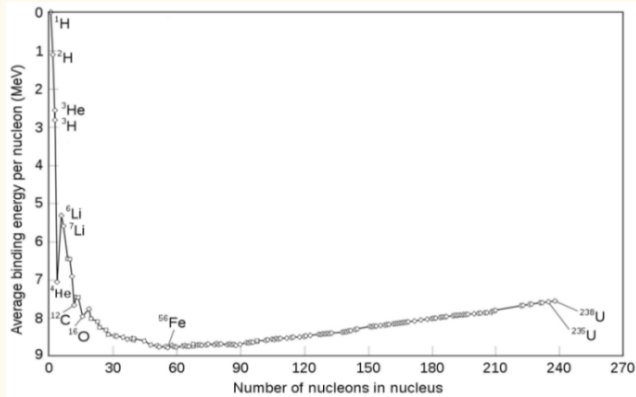
$$931.5 \frac{\text{MeV}}{\text{amu}}$$

# Binding energy is used to compare atomic stability, reaction energy, probability of fission/fusion



**Figure 6.** Binding energy curve

# Light nuclei get more stable by fusion; heavy nuclei get more stable by fission



**Figure 7.** Inverted binding energy curve



**Q-value**

# The Q-value for a reaction indicates energy requirements



- (1) Conservation of nucleons
- (2) Conservation of charge
- (3) Conservation of momentum
- (4) Conservation of energy

# Compute the Q value for this reaction



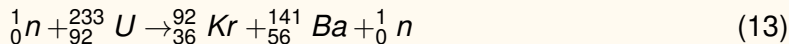
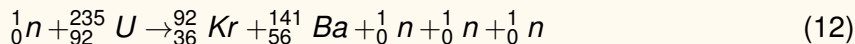
$$Q = [(M_{\text{Li}} + M_n) - (M_{\alpha} + M_H)]c^2 \quad (10)$$

$$Q = 4.78 \text{ MeV} \quad (11)$$

$$Q > 0 \rightarrow ?$$

$$Q < 0 \rightarrow ?$$

# What is the energy of fission from thorium and uranium fuel cycles?



For comparison, burning a carbon atom releases 4 eV

# Radioactivity

# Scientists in late 1800s made discoveries which would change the course of science, medicine, and history in the 20th Century



The Nobel Prize in Physics 1903 was divided, one half awarded to Antoine Henri Becquerel *"in recognition of the extraordinary services he has rendered by his discovery of spontaneous radioactivity"* the other half jointly to Pierre Curie and Marie Curie, née Skłodowska *"in recognition of the extraordinary services they have rendered by their joint researches on the radiation phenomena discovered by Professor Henri Becquerel"*.

**Figure 8.** 1903 Nobel Prize in physics winners

# Radiation is energy as particles or electromagnetic waves, like sunshine

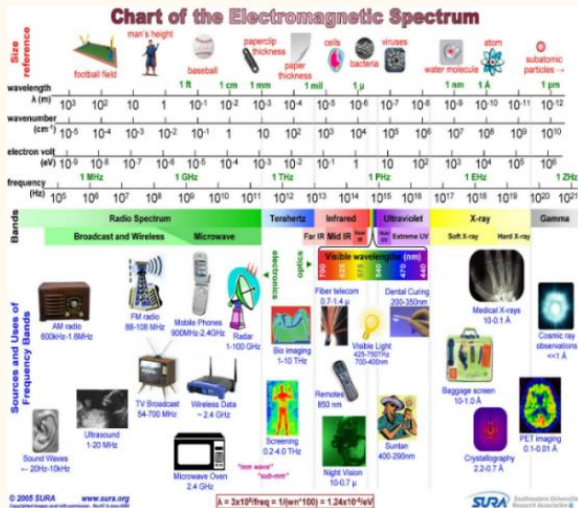
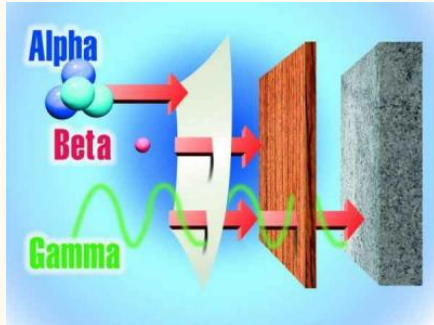
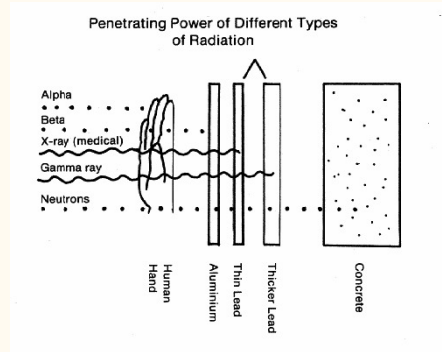


Figure 9. Chart of the electromagnetic spectrum

# Radiation is energy as particles or electromagnetic waves



**Figure 10.** Radiation shielding materials



**Figure 11.** Radiation shielding materials



# Radiation is energy as particles or electromagnetic waves

Charged particles are –

- (1) protons (+)
- (2) alpha (++)
- (3) beta (+/-)
- (4) heavy ions (varying)
- (5) U, Pu (3+, 4+, 5+, 6+)

Neutrons have no charge, but are also ionizing

Ionizing rays are x-rays, g-rays, cosmic rays

# Ernest Rutherford discovered the three main kinds of radioactive decay (Chemistry Nobel 08)

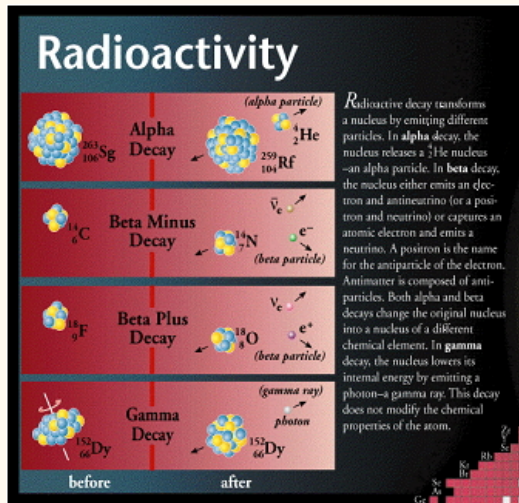


Figure 12. Types of radioactive decay

# Alpha particle decay is the ejection of a helium atom from a heavy nucleus



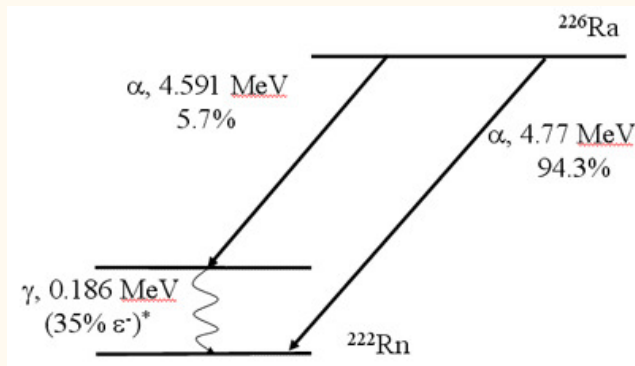
$$Q = 4.268 \text{ MeV} \quad (15)$$

Sometimes nucleus decays to excited state and then emits a gamma ray (for any kind of radiation)

Discrete energy spectrum

Some % decay at certain energy

# Radium-226 decays to radon-222



**Figure 13.** Radium-226 decay

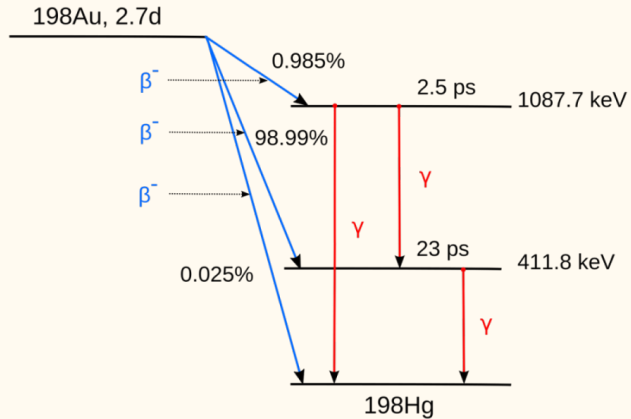
## $\beta^-$ particle decay occurs for unstable nuclei with excessive neutrons



The neutron is converted to a proton and a beta negative (electron) and anti-neutrino (momentum) are ejected

Sometimes nucleus decays to excited state and then emits a gamma ray (for any kind of radiation)

# Gold-198 decays to mercury-198



**Figure 14.** Gold-198 decay

## $\beta^+$ particle decay occurs for unstable nuclei with deficient neutrons



The proton is converted to a neutron and a beta positive (positron) and neutrino (momentum) are ejected

Sometimes nucleus decays to excited state and then emits a gamma ray (for any kind of radiation)

# Sodium-22 decays to neon-22

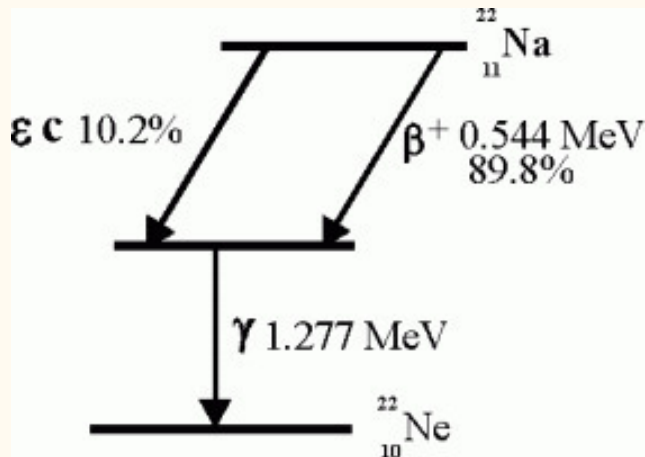
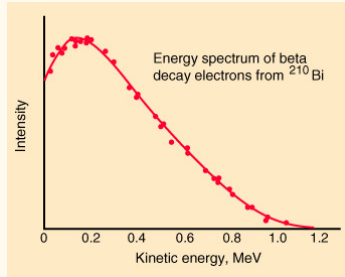


Figure 15. Sodium-22 decay



**In both forms of beta decay, the energies are exhibited by a continuous energy spectrum**



**Figure 16.** Beta decay energy spectrum

$$\overline{E}(\beta^-) = 0.3E_{MAX} \quad (18)$$

$$\overline{E}(\beta^+) = 0.4E_{MAX} \quad (19)$$

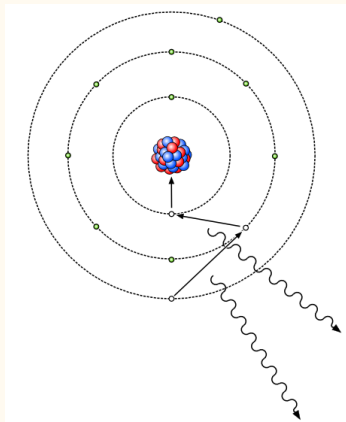
# Electron capture also occurs for unstable nuclei with deficient neutrons



The K shell electron (not free) is absorbed by the nucleus to convert the proton to a neutron

Sometimes nucleus decays to excited state and then emits a gamma ray (for any kind of radiation)

# Electron is captured by the nucleus



**Figure 17.** Notional electron capture

# Electron capture and $\beta^+$ are competing processes



$$Q = 2.842 \text{ MeV} \quad (23)$$

$Q > 1.022 \text{ MeV}$  positron favored

$Q < 1.022 \text{ MeV}$  electron capture favored

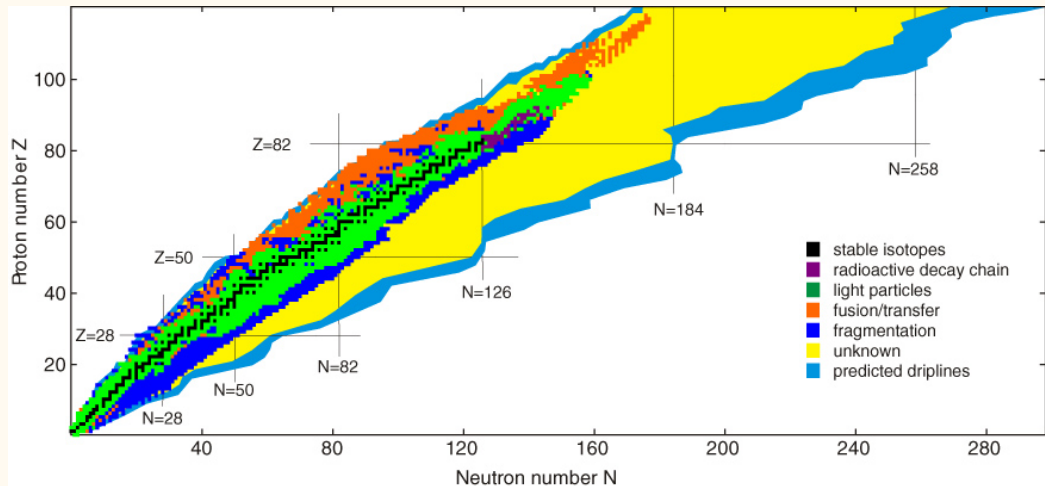
Why?

# $\gamma$ -ray emission is not a decay mode, but superfast de-excitation of the nucleus

Interactions of gamma rays with matter is a very important branch of nuclear physics

# Chart of the nuclides

# Chart of the nuclides wide shot



**Figure 18.** Chart of the nuclides

# Chart of the nuclide example

|                                                                                                  |                                                                                      |                                                                                                                      |
|--------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Ra 224</b><br>3.66 d<br>$\alpha$ 5.6854<br>5.4486...<br>$\gamma$ 241..., C14<br>$\sigma$ 12.0 | <b>Ra 225</b><br>14.8 d<br>$\beta^-$ 0.3, 0.4<br>$\gamma$ 40<br>$e^-$                | <b>Ra 226</b><br>1600 a<br>$\alpha$ 4.7843, 4.601...<br>$\gamma$ 186...<br>C14<br>$\sigma$ 12.8<br>$\sigma_T < 5E-5$ |
| <b>Fr 223</b><br>21.8 m<br>$\beta^-$ 1.1...<br>$\alpha$ 5.34<br>$\gamma$ 50, 80, 235...          | <b>Fr 224</b><br>3.3 m<br>$\beta^-$ 2.6, 2.8...<br>$\gamma$ 216, 132, 837<br>1341... | <b>Fr 225</b><br>4.0 m<br>$\beta^-$ 1.6...<br>$\gamma$ 182, 32, 225<br>200...                                        |
| <b>Rn 222</b><br>3.825 d<br>$\alpha$ 5.48948...<br>$\gamma$ (510)<br>$\sigma$ 0.74               | <b>Rn 223</b><br>23.2 m<br>$\beta^-$<br>$\gamma$ 593, 417, 636<br>655...             | <b>Rn 224</b><br>1.78 h<br>$\beta^-$<br>$\gamma$ 261, 266...                                                         |

**Figure 19.** Chart of the nuclides example



# Decay chains

# Thorium decays to lead

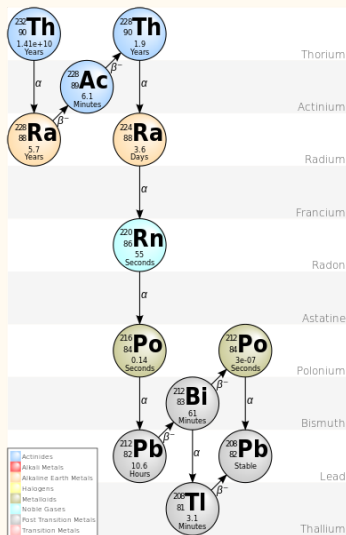


Figure 20. Thorium decay chain

# Neptunium decays to thallium

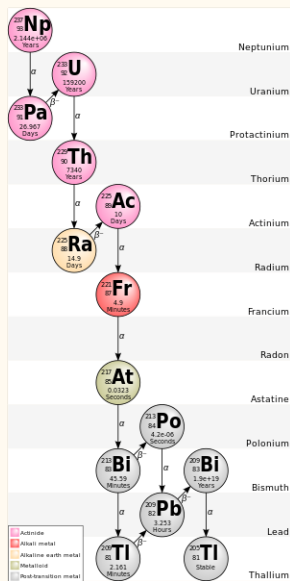
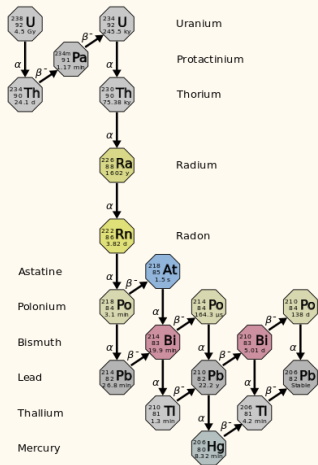


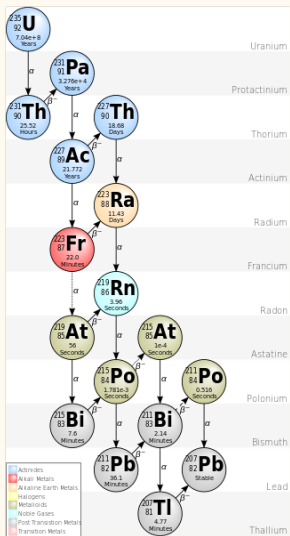
Figure 21. Neptunium decay chain

# Uranium-238 decays to lead



**Figure 22. Uranium-238 decay chain**

# Uranium-235 decays to lead



**Figure 23. Uranium-235 decay chain**

## Half life

# A radioactive isotope decays with a unique characteristic time

Decay is stochastic, characterized by Poisson distribution

So the decay of any one radionuclide cannot be predicted

The probability per time that a nucleus will decay is a constant

# Decay is described by a Poisson process

## Poisson process notes

Occurrences are randomly distributed in time

What is the 'occurrence' here?

Homogeneous, long term decay rate is constant

$$\mu \equiv \lambda \Delta t \quad (24)$$

$$P(n) = e^{-\mu} \cdot \frac{\mu^n}{n!} \quad (25)$$

Defintion of expected value and variance on the OER



# A radioactive isotope decays with a unique characteristic time

$n(t)$  atoms at  $t$  have decayed in  $dt$

$$\lambda n(t)dt \quad (26)$$

The number of atoms that decay on average in the interval  $t, (t + dt)$  can be expressed as –

$$-dn(t) = \lambda n(t)dt \quad (27)$$

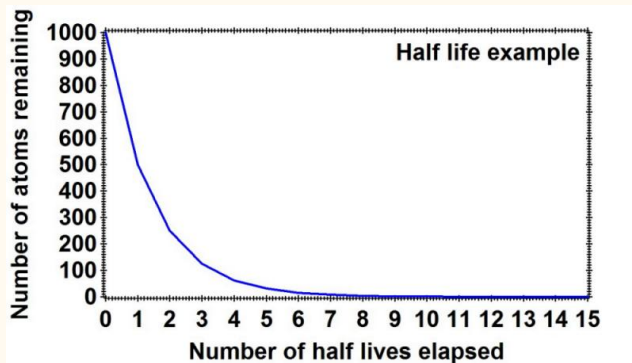
You should be able to solve this for  $n(0) = n_0$

# Radionuclides are characterized by half life

$$t_{\frac{1}{2}} = \frac{\ln 2}{\lambda} \quad (28)$$

Half life can be derived from the decay law

# A radionuclide is 'negligible' after 10 half-lives have elapsed



**Figure 24.** Notional half life

Why would this be important?

# How was half-life discovered?

Rutherford discovered alpha and beta particle decay

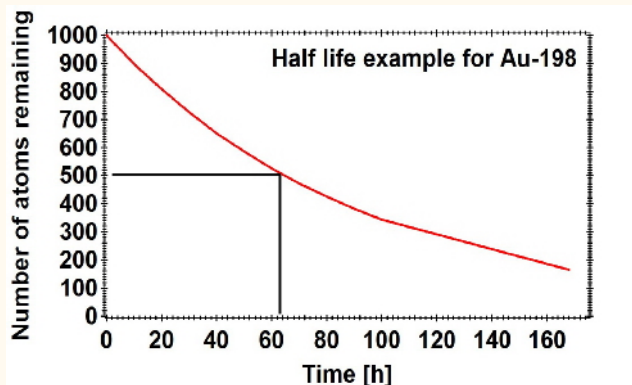
In the course of, he noticed that there was a characteristic time

Basically observed decay time was the same for each atom

Experiments with thorium led to the discovery of radon

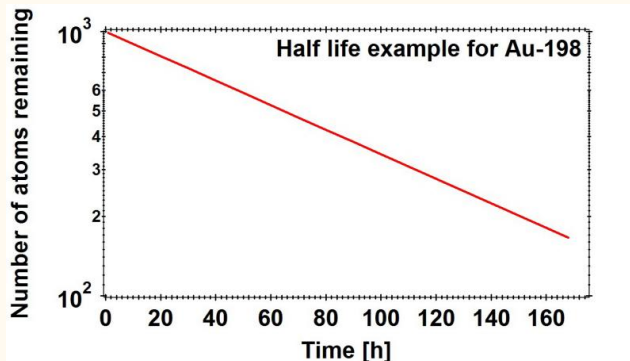
Then observing the radon gas led to the decay/half-life (65 s)

# Gold-198 graphical half life



**Figure 25.** Gold-198 half life

# Engineers like straight lines



**Figure 26.** Gold-198 half life on a log scale

Slope of the line is the decay constant



# Acronyms I

**OER** Online Educational Resource.

**UI** University of Idaho.

**UIIF** Idaho Falls Center for Higher Education.